Du Pei

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Employment

- Associate Professor, 2022-current.
  - Centre for Quantum Mathematics, University of Southern Denmark.
- Postdoctoral Fellow, 2020–2022.
  - Center for Mathematical Sciences and Applications, Harvard University.
- Postdoctoral Fellow, Fall 2019.
  - Mathematical Sciences Research Institute.
- Visiting Scholar, Fall 2019.
  - Department of Mathematics, UC Berkeley.
- QGM-Caltech Joint Postdoc, 2016–2019.
  - Center for Quantum Geometry of Moduli Spaces, Aarhus University.
  - Walter Burke Institute for Theoretical Physics, California Institute of Technology.

Future Visiting Position

- Chern Professor, Simons Laufer Mathematical Sciences Institute, Fall 2026.
- Sabbatical member, Institute for Advanced Study, Spring 2027.

Education

- California Institute of Technology, Ph.D. in Physics, 2011–2016.
- Peking University, B.Sc. in Physics, 2007–2011.

Research Interest

I use ideas motivated by quantum physics to discover new structures and tackle old problems in mathematics. At the same time, I apply advanced mathematical tools to better understand algebraic, geometric, topological and categorical structures in quantum field theories. Some of my specific interests include:

- topological quantum field theory
- quantum phases of matter
- geometry of moduli spaces
- new invariants of 3- and 4-manifolds
- categorification and quantization

Grants & Awards

2024 Co-PI for Innovation Fund Denmark grant: "*Topological Photonic Quantum Computing and its Applications.*"

2023 Co-Investigator for Simons Collaboration: "*New structures in low-dimensional topology.*"

2022 Villum Young Investigator grant: "*New Invariants in Low-Dimensional Topology via Quantum Field Theory.*"

2014 GEAR travel award for Graduate Internship, from NSF grants: *"RNMS: Geometric Structures and Representation Varieties."*

Publications & Preprints

- **"Gauge theory and skein modules."** arXiv:2601.16213 [hep-th].
- **"Generalized symmetry of T[M] theories, II."** *Journal of High Energy Physics,* 05 (2026) 087, arXiv:2511.13696 [hep-th] (with S. Gukov, P. S. Hsin)
- **"E-strings and Four-Manifolds,"** arXiv:2511.11807 [hep-th] (with D. H. Wu)
- **"A new approach to (3+1)-dimensional TQFTs via topological modular forms,"** arXiv:2509.12402 [math.AT] (with S. Gukov, V. Krushkal, and L. Meier)
- **"On mixed't Hooft anomalies of emergent symmetries,"** *Journal of High Energy Physics,* 07 (2026) 195 arXiv:2506.06432 [hep-th] (with W. Gu and X. Yu)
- **"Branes and DAHA Representations,"** Springer (2023) arXiv:2206.03565 [hep-th] (with S. Gukov, P. Koroteev, S. Nawata, and I. Saberi,)
- **"Holomorphic CFTs and topological modular forms,"** *Commun. Math. Phys.* **401**, 325–332 (2023). arXiv:2112.10724 [hep-th] (with Y.-H. Lin)
- **"Symmetries of 2d TQFTs and Equivariant Verlinde Formulae for General Groups,"** arXiv:2111.08032 [hep-th] (with S. Gukov, C. Reid, and A. Shehper)
- **"On Phases of 3d N=2 Chern-Simons-Matter Theories,"** *Nuclear Physics B, 973, (2021)* arXiv:2105.02247 [hep-th] (with W. Gu, and M. Zhang,)
- **"Generalized Symmetries of $T[M]$ Theories, I,"** *Journal of High Energy Physics,* (2021) 232. arXiv:2010.15890 [hep-th] (with S. Gukov, and P.-S. Hsin)
- **"Rozansky–Witten geometry of Coulomb branches and logarithmic knot invariants,"** *Journal of Geometry and Physics,* Volume 168, October 2021, 104311. arXiv:2005.05347 [hep-th] (with S. Gukov, P.-S. Hsin, H. Nakajima, S. Park, and N. Sopenko)
- **"Trialities of Minimally Supersymmetric 2d Gauge Theories,"** *Journal of High Energy Physics,* (2020) 79. arXiv:1910.13455 [hep-th] (with S. Gukov and P. Putrov)
- **"4-manifolds and topological modular forms,"** *Journal of High Energy Physics,* (2021) 84. arXiv:1811.07884 [hep-th], (with S. Gukov, P. Putrov and C. Vafa)
- **"Identities for Poincaré polynomials via Kostant cascades,"***Journal of Lie Theory,* **30**, No. 4, 1131–1160, (2020) arXiv:1810:05615 [math.CO] (with J. E. Andersen and J. C. Jantzen)
- **"3d TQFTs from Argyres–Douglas theories,"***Journal of Physics A*, **53**, No. 43, (2020), arXiv:1809.04638 [hep-th] (with M. Dedushenko, S. Gukov, H. Nakajima and K. Ye)

- **"Mirror symmetry with branes by equivariant Verlinde formulae,"** *Geometry and Physics: A Festschrift in honour of Nigel Hitchin*, Volume **1**, Chapter 9, arXiv:1712.04408 [math.AG] (with T. Hausel and A. Mellit)
- **"Argyres–Douglas theories, chiral algebras and wild Hitchin characters,"** *Journal of High Energy Physics*, **1** (2018): 150. arXiv:1701.08782 [hep-th] (with L. Fredrickson, W. Yan and K. Ye)
- **"BPS spectra and 3-manifold invariants,"** *J. Knot Theor. Ramifications* **29** 02, 2040003 (2020) arXiv:1701.06567 [hep-th]. (with S. Gukov, P. Putrov and C. Vafa)
- **"The Verlinde formula for Higgs bundles,"** arXiv:1608:01761 [math.AG]. (with J. E. Andersen and S. Gukov)
- **"Equivariant Verlinde algebra from superconformal index and Argyres–Seiberg duality,"** *Communications in Mathematical Physics*, **357**.3 (2018): 1215-1251. arXiv:1605.06528 [hep-th]. (with S. Gukov, W. Yan and K. Ye)
- **"A 3d-3d appetizer,"** *Journal of High Energy Physics*, **11** (2016): 8. arXiv:1503.04809 [hep-th]. (with K. Ye)
- **"Equivariant Verlinde formula from fivebranes and vortices,"** *Communications in Mathematical Physics*, **355**.1 (2017): 1-50. arXiv:1501.01310 [hep-th] (with S. Gukov)

Lecture Series

07/2025 *Mini-course "On the space of quantum field theories,"* Yau Mathematical Science Center, Tsinghua University, Beijing, China.

08/2024 *Mini-course on "Quantum field theory and topological modular forms,"* Summer School at Southeastern University, Nanjing, China.

10/2022 *Mini-course on "VOAs from higher-dimensional QFTs,"* School and workshop on "Quantum symmetries: Tensor categories, Topological quantum field theories, Vertex algebras", CRM, Montreal.

08/2022 *Mini-course on "Invariants of Manifolds from 6d QFTs,"* School on Physical Mathematics of Quantum Field Theory, UMass Amherst.

06/2022 *Mini-course on "Mirror symmetry and Higgs moduli spaces,"* Workshop-School on Mirror Symmetry and Moduli Spaces, Lisbon.

11/2020 *Mini-course on "TQFTs from Coulomb Branches,"* School–Conference on Vertex Algebras and Tensor Categories, Higher School of Economics, Moscow/Zoom.

10/2019 *Mini-course on "Invariants of Manifolds from Non-Gauge Theories,"* School and Workshop on Gauge Theories and Differential Invariants, ICTP, Trieste.

08/2019 *Mini-course on "Physics and geometry of BPS states,"* Introductory Workshop: Holomorphic Differentials in Mathematics and Physics, MSRI.

07/2018 *Mini-course on "Verlinde formula for complex Chern–Simons theory,"* Summer School on Geometry, Quantum Topology and Asymptotics, Geneva.

10/2015 *Mini-course on "New TQFTs and equivariant integration over the moduli space of Higgs bundles"*, NS@50, Chennai Mathematical Institute.

Selected Conference Talks

09/2026 *"6d theories and 4-manifolds,"* Workshop on Gauge Theory, Topology, and Physics, OIST, Okinawa, Japan.

08/2026 *"On the classification of 2d TQFTs with time-reversal symmetry,"* Physics and Mathematics of Generalized Symmetry in QFT, Kavli IPMU, Japan.

08/2026 *"On classifications of defects in quantum field theories,"* Workshop on Generalized Symmetries in HEP and CMP 2026, Institute of Physics, Chinese Academy of Sciences, Beijing, China.

08/2026 *"Gauge theory and skein modules,"* International Congress of Basic Science, BIMSA, Beijing, China (invited lecture).

07/2026 *"On boundary conditions of quantum field theories,"* Elliptic objects, von Neumann algebras, and Functorial Field Theory, Bernoulli Center, EPFL, Switzerland.

06/2026 *"Gauge theory and skein modules,"* String–Math 2026, Edinburgh, United Kingdom.

09/2025 *"Gauge theory and skein modules,"* Conference on Gauge summer with BV, Scalea, Italy.

08/2025 *"Gauge theory and skein modules,"* ReNewQuantum concluding conference, SDU, Odense, Denmark.

07/2025 *"On boundary conditions and phases of quantum field theories,"* International Congress of Basic Science, BIMSA, Beijing, China (listed as a plenary lecture in the Formal Quantum Field Theory track).

06/2024 *"Topological modular forms and phases of quantum field theories,"* BPS Dynamics and Quantum Mathematics, Galileo Galilei Institute, Florence, Italy.

01/2024 *"On new invariants and phases of supersymmetric quantum field theories,"* Workshop on String Theory and Quantum Field Theory 2024, Fudan University, Shanghai, China.

08/2023 *"Brane quantization and DAHA representations,"* workshop on Mathematics Inspired by String Theory, CUHK, Hong Kong.

08/2022 *"Quantum Field Theory and Topological Modular Forms,"* Simons Summer Workshop, SCGP.

07/2022 *"Wall-Crossing from Higgs Bundles to Vortices,"* Conference on Enumerative Invariants, Quantum Fields and String Theory Correspondences, Institut Mittag-Leffler.

11/2021 *"TQFTs from Coulomb Branches,"* Banff workshop on Quantum Field Theories and Quantum Topology Beyond Semisimplicity.

07/2021 *"New Invariants of Manifolds from String Theory,"* Simons Summer Workshop.

07/2021 *"Verlinde Formula for PSL(2,C) Higgs bundles",* VBAC online workshop on Geometry and Quantization of Moduli Spaces

06/2021 *"Verlinde Formula, Brane Quantization, and Mirror Symmetry,"* School on Mirror Symmetry and Moduli Spaces 2021.

01/2021 *"Quantization of Coulomb Branches and Double Affine Hecke Algebras,"* JMM Special Session on Quantum Algebra and Geometry.

11/2020 *"Modularities from fivebranes,"* KITP program on Modularity in Quantum Systems, Santa Barbara/Zoom.

10/2020 *"Branes in the moduli space of Higgs bundles,"* Workshop on Special holonomy and branes, American Institute of Mathematics, San Jose/Zoom.

07/2020 *"On compatifications of 6d relative theories,"* String–Math 2020, South Africa/Zoom.

02/2020 *"BPS states and 3d TQFTs,"* Workshop on Topological and Geometric Recursion, Institute of the Mathematical Sciences of the Americas, Miami.

10/2019 *"4d symplectic duality,"* Workshop on 3d Mirror Symmetry and AGT Conjecture, IASM, Zhejiang University.

09/2019 *"4-manifolds and topological modular forms,"* Workshop on 4-manifolds, Max Planck Institute for Mathematics, Bonn.

06/2019 *"3d TQFTs from 4d CFTs,"* Workshop on Mathematics and Physics of Knots, Mittag-Leffler Institute.

06/2019 *"Verlinde algebras for Higgs bundles,"* Conference on Verlinde Algebras and Grassmannians, Sun-Yat Sen University.

06/2019 *"New TQFTs from DAHA (Part 2),"* Conference on Geometric Representation Theory and Low-Dimensional Topology, ICMS Edinburgh.

06/2019 *"Taming the Non-Unitary Zoo with Wild Higgs Bundles,"* New Developments in Quantum Topology, Berkeley.

06/2019 *"Introduction to q-series $\hat{Z}_a(M_3; q)$,"* New Developments in Quantum Topology, Berkeley.

03/2019 *"3d TQFTs from non-Lagrangian theories,"* Workshop on Higher Symmetries: Theory and Applications, Aspen Center for Physics.

03/2019 *"4-manifolds and topological modular forms,"* Conference on Hidden Algebraic Structures in Topology, Caltech.

03/2019 *"Practical computation for q-series $\hat{Z}_a$,"* workshop on Modularity and 3-manifolds, ICERM, Providence.

02/2019 *"Modular tensor categories from wild Higgs bundles,"* Workshop on Holomorphic Differentials in Mathematics and Physics, Simons Center for Geometry and Physics.

02/2019 *"Non-unitary modular categories from the Coulomb branch,"* Workshop on Representation theory, gauge theory, and integrable systems, Kavli IPMU

12/2018 *"Modular tensor category from the Coulomb branch,"* Workshop on Gauge Theory and Vertex Algebra, Simons Center for Geometry and Physics.

11/2018 *"On new homological and homotopy-theoretic invariants of manifolds,"* Program on Quantum Knot Invariants and Supersymmetric Gauge Theories, KITP.

09/2018 *"Modular tensor categories from wild Higgs bundles,"* Workshop on Geometric and Categorical Aspects of CFTs, Oaxaca, Mexico.

09/2018 *"Wild Higgs bundles and modular categories,"* Workshop on Geometry and Physics of Quantum Curves, Banff, Canada.

07/2018 *"On mirror symmetry of (B, A, A)–branes,"* Conference on Quantum Fields, Geometry and Representation Theory, ICTS, Bangalore.

07/2018 *"2d TQFTs labeled by three-manifolds,"* CQ3MI Summer School and Workshop, University of Southern California.

06/2018 *"On mirror symmetry of (B, A, A)–branes,"* Conference on Vector Bundles on Algebraic Curves 2018: Gauge Theory and Complex Geometry, CIRM, Luminy.

04/2018 *"On categorification of the WRT invariant,"* Workshop on Categorification in Mathematical Physics, Simons Center for Geometry and Physics.

03/2018 *"2d TQFTs from 3-manifolds,"* Conference on Quantum Knot Homology and Supersymmetric Gauge Theories, Aspen Center for Physics.

01/2018 *"BPS spectra and invariants for 3- and 4-manifolds,"* Winter School on Partition Functions and Automorphic Forms, JINR.

01/2017 *"Wild Hitchin characters,"* workshop on Quantization and Moduli Spaces, University of Luxembourg.

09/2016 *"The Verlinde formula for Higgs bundles,"* Hitchin70: Differential Geometry and Quantization, Aarhus University.

02/2016 *"Verlinde formula for Higgs bundles,"* Workshop on Higgs Bundles in Geometry and Physics, IWH Heidelberg.

10/2015 *"Branes and mirror symmetry in the Hitchin moduli space,"* NS@50: Fifty Years of the Narasimhan–Seshadri Theorem, Chennai Mathematical Institute.

11/2014 *"Equivariant integration over the Hitchin moduli space"*, Gauged Sigma-Models in Two Dimensions, Simons Center of Geometry and Physics.

05/2014 *"TQFT and the geometry of moduli spaces"*, GEAR Junior Retreat, University of Michigan.

Colloquia

05/2024 *"Hidden algebraic structures in geometry from quantum field theory,"* Geometry and Physics Colloquia, Uppsala University, Sweden.

07/2022 *"On Hidden Mathematical Structures from Quantum Field Theories,"* Johannes Gutenberg Universität Mainz, Germany.

05/2021 *"Hidden Algebraic Structures in Geometry from Fivebranes,"* Pure Maths Colloquium, University of Sheffield.

02/2021 *"On Quantization of Coulomb Branches,"* Mathematical colloquium, BMSTU Moscow.

12/2020 *"Hidden algebraic structures in geometry from fivebranes,"* Mathematics Colloquium, UCLA.

02/2020 *"New developments in physical mathematics,"* University of Amsterdam.

04/2019 *"New developments in physical mathematics,"* Humboldt University of Berlin.

11/2018 *"4-manifolds and topological modular forms,"* Max Planck Institute for Mathematics, Bonn.

Selected Seminars

05/2026 *"Gauge theory and skein modules,"* Algebra, Geometry and Topology Seminar, ISTA, Klosterneuburg, Austria.

12/2025 *"On moduli spaces of $T[M]$ theories,"* Yau Center, Southeast University, online.

04/2024 *"Informal Seminar about Generalized Symmetries and Dimensional Reduction,"* Geometry and Physics seminar series, Uppsala University, Sweden.

03/2024 *"On new invariants and phases of supersymmetric quantum field theories,"* Center for Quantum and Topological Systems, NYU Abu Dhabi.

11/2022 *"Conformal field theories and topological modular forms,"* Particle Theory Seminar, Cornell University.

05/2021 *"On Phases of 3d N=2 Chern-Simons-Matter Theories,"* Caltech High Energy Theory Journal Club.

11/2020 *"Polarizations and symmetries of $T[M]$ theories,"* Oxford String Theory Journal Club.

11/2020 *"Integrability from fivebranes,"* Geometry and Mathematical Physics Seminar, SISSA, Italy.

11/2019 *"On new homological and homotopy-theoretic invariants of manifolds,"* String-Math Seminar, UC Berkeley.

05/2019 *"Brane Quantization and Representations of DAHA,"* iGAP Seminar, ICTP, Trieste.

02/2019 *"Modular tensor categories from the Coulomb branch,"* Symplectic Geometry, Gauge Theory, and Categorification Seminar, Columbia University.

09/2018 *"Higgs bundles, branes and mirror symmetry,"* Séminaire de la Tortue, University of Geneva.

Organizational Activities

- Special session on "Invariants of manifolds from quantum field theory and string theory" at 29th Nordic Congress of Mathematicians, 2023.
- Workshop on "Generalized Cohomology and Physics," ICTP, Trieste, 2021.
- Colloquium Series at CMSA, Harvard, Fall 2021–present.
- Seminar series on "Quantum Matter in Mathematics and Physics" at CMSA Harvard, 2020–2021.
- Mathematical Physics seminar at CMSA Harvard, Summer 2020.

Professional Service

- Editor for *Advances in Theoretical and Mathematical Physics*.
- Referee for *Journal of Differential Geometry*, *Communications in Mathematical Physics*, *Letters in Mathematical Physics*, *Reviews in Mathematical Physics*, *Journal of High Energy Physics*, etc.